Alex I.

SOFTWARE ENGINEER

Education
Computer Science and Software Engineering

Language proficiency
English – B2+
Polish – B1
Swedish – A2

Domains
Telecommunications
Cloud Storage Systems
Library Systems
Oil and Gas Industry
UAV Systems
Aerospace Systems

Software engineer with 14+ years of commercial experience.
I possess a strong background in hardware and software development, with a primary focus on C/C++ programming and Linux-based systems. My experience includes backend development, distributed service communication, persistent data storage, and integration with object storage platforms such as AWS S3 and Ceph RGW. I also have extensive experience developing MATLAB-based algorithms and applications for image analysis.

Programming Languages
C++, C, C#, Python, Java, JavaScript, Ruby, MATLAB, Scala, HTML.

Operating Systems
Linux, Embedded Linux.

Tools
Kubernetes, Yocto Project, Grafana, Prometheus, Ceph RADOS, Ceph CSI, Qt Designer, Qt Designer .ui forms, RBD, System Tool Kit, systemd, Rook Operator, Visual Assist, CMake, qmake, gcc, GDB, Valgrind, Qt Test, Pytest, Visual Studio, Qt Creator.

Frameworks & Libraries
GStreamer, Qt, Qt Widgets, QSS, Ruby on Rails, .NET, Node.js, WinForms, LevelDB, V4L2, jQuery, Qt DBus, Qt WebSockets.

Multimedia
WebRTC, Qt Multimedia, Linphone-SDK, liblinphone, mediasoup, libmediasoupclient.

Databases
MySQL, PostgreSQL, SQLite, LevelDB, Microsoft SQL Server.

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Network
SDP, Nginx, SIP, RTP, SNMP, DHCP, TFTP.

Cloud
Ceph RGW, S3 API, AWS (S3).

Embedded
BitBake, NXP i.MX.

Development Processes
Jira, Asana.

Source control systems
Git, GitLab, Bitbucket.

Architecture
Qt Model/View.

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Projects

SIP DESK PHONE PLATFORM FOR AUDIO AND VIDEO COMMUNICATION

A SIP desk phone platform for audio and video communication, supporting both camera-enabled and audio-only device models. The project included the native phone application, web management interface, automated provisioning tools, and embedded system customization. It covered everyday calling workflows such as SIP registration, incoming and outgoing calls, video calls, call hold, transfer, conferencing, phonebook access, direct IP dialing, remote configuration, and device setup.

Project roles
Software Engineer

Period
05.2024 – till now

Responsibilities

- Refactored SIP call handling toward a cleaner MVC/MVVM-style architecture by separating UI rendering from call state, user actions, media negotiation, and Linphone-SDK integration logic;
- Incrementally decomposed legacy Qt Widgets call-flow code into clearer domain, presentation, and SDK-integration boundaries while preserving existing UI behavior and QObject signal compatibility;
- Hardened video-call and conference UI behavior for edge cases such as held calls, second-line calls, remote hangups, stale overlays, and active video stream switching;
- Extended and customized Linphone-SDK (liblinphone) backend logic to implement SIP failover and backup-server connectivity for improved device reliability;
- Developed WebRTC video-conferencing UI and signaling flows: conference creation/join, incoming invite screens, participant roster, SIP/IP/phonebook add dialogs, status strip, control panel, video tile layout, and audio/video device controls;
- Autoprovisioning tooling: migrated a WinForms-based configuration utility from .NET Framework 4.8 to .NET 8, unified project file formats across the device ecosystem, and implemented provisioning workflows using SNMP, DHCP, TFTP, and SQLite;
- Collaborated on Yocto/BitBake customization, i.MX embedded Linux target integration, and device-side service configuration for hardware-software integration.

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Environment

C++17, C#, JavaScript, Embedded Linux, Yocto, Qt Designer, Qt Designer .ui forms, systemd, CMake, qmake, GCC, gdb, Valgrind, QtTest, Qt Creator, GStreamer, Qt 5.15.7, Qt Widgets, QSS, .NET 8, Node.js, WinForms, V4L2, Qt DBus, Qt WebSockets, WebRTC, Qt Multimedia, Linphone-SDK, liblinphone, mediasoup, libmediasoupclient, SQLite, SDP, SIP, RTP, SNMP, DHCP, TFTP, BitBake, NXP i.MX, Asana, Git, GitLab, Qt Model/View.

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REMOTE SENSING SATELLITES DATABASE

A web-based platform for managing Earth observation satellite information, research datasets, and large-scale remote sensing archives. Ruby on Rails powered the web interface and application backend, including catalogue management, search, and metadata workflows. Performance-critical file ingestion and data processing were handled by native C++ components, responsible for efficient streaming, validation, and preparation of large satellite imagery and telemetry archives. PostgreSQL stored structured metadata, while Ceph provided distributed objects and persistent storage. The infrastructure was deployed in Kubernetes environments using Rook Operator and Ceph CSI, with Python automation supporting storage validation and monitoring.

Project roles
Software Engineer

Period
04.2022 – 04.2024

Responsibilities

- Developed Ruby on Rails web interfaces and workflows for satellite catalogues, metadata, dataset search, and archive access;
- Developed C++ ingestion components with POSIX streaming I/O and optimized memory buffers for large imagery and telemetry archives;
- Implemented multithreaded data validation, checksum calculation, and file preparation for bulk ingestion;
- Integrated Ceph RGW via S3 API for satellite data storage and retrieval, including pre-signed URLs for secure file access;
- Deployed and maintained Ceph services in Kubernetes using Rook Operator;
- Configured Ceph CSI for RBD, enabling dynamic persistent volume provisioning;
- Investigated Ceph performance issues, analyzing I/O latency, throughput, OSD behavior, and BlueStore utilization;
- Developed Python scripts for storage health checks, deployment validation, data integrity, and performance testing;
- Monitored storage infrastructure with Prometheus and Grafana, identifying bottlenecks and resource issues.

Environment
C++, Python, Ruby, Kubernetes, Grafana, Prometheus, Ceph RADOS, Ceph CSI, RBD, Rook Operator, pytest, Ruby on Rails, PostgreSQL, Ceph RGW, S3 API, Git.

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SOFTWARE MODULES FOR UAV

Developed software modules for digital image processing systems used in UAV optical sensors. The project combined MATLAB-based algorithm development and analysis with C++ implementation of performance-critical image processing components. Key functionality included raw sensor data conversion, image stabilization, format conversion, and multi-sensor image fusion to improve image quality and processing efficiency.

Project roles
Software Engineer

Period
03.2020 – 03.2022

Responsibilities

- Developed digital image processing modules for UAV optical sensor systems, combining MATLAB algorithm development with C++ implementation of performance-critical components;
- Prototyped and evaluated debayering algorithms in MATLAB and implemented C++ processing modules to convert raw Bayer sensor data into full-color images;
- Developed C++ image format conversion and preprocessing components to support different sensor output formats and ensure compatibility across processing pipelines;
- Optimized image frame processing in C++, focusing on efficient memory buffer management and reducing unnecessary data copying during image operations;
- Developed and evaluated image stabilization algorithms using MATLAB, implementing selected processing operations in C++ for integration into image processing workflows;
- Designed and validated multi-sensor image fusion algorithms in MATLAB and integrated the resulting processing modules into the software pipeline;
- Tested image processing modules using representative sensor datasets, analyzing output quality, processing accuracy, and performance.

Environment
C++, MATLAB, PostgreSQL, Git, Bitbucket.

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SOFTWARE FOR PROCESSING AND ANALYSIS OF SATELLITE EARTH OBSERVATION IMAGES

The software includes a range of modules with functions such as determining linear resolution, enhancing image quality through prediction of image type and other factors, stitching frames taken by a remote sensing satellite, eliminating optical distortions between frames. Additionally, one of the key components of the project is the development of a compression algorithm based on Adaptive differential pulse-code modulation (ADPCM).

These features are aimed at providing a comprehensive toolkit for working with satellite imagery, enabling precise resolution analysis, effective data compression, and optimized image processing workflows for various remote sensing applications.

Project roles
Software Engineer

Period
01.2018 – 02.2020

Responsibilities

- Developed algorithms for standardizing brightness levels across multiple frames and eliminating optical defects, improving the overall quality and consistency of satellite imagery;
- Designed and implemented software to analyze the linear resolution of satellite images using the sharp edge method, enhancing the accuracy of imaging performance evaluation;
- Created algorithms to assess image sharpness and edge clarity, ensuring precise resolution measurements for remote sensing applications and contributing to reliable data interpretation;
- Developed software to predict quantization ranges for remote sensing satellites by analyzing Earth observation data and estimating the albedo range, improving sensor calibration and minimizing data loss;
- Automated the prediction of quantization ranges, streamlining workflows, and improving operational efficiency;
- Designed and implemented a custom image compression algorithm using ADPCM with multiple quantizers, achieving optimized compression ratios and enhanced flexibility for different data sizes.

Environment
C++, MATLAB, System Tool Kit, CMake, Visual Studio, PostgreSQL, Jira, Git.

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LIBRARY SYSTEM

A library management platform serving academic, municipal, and national library clients. The system maintained large bibliographic catalogues for physical collections alongside metadata and digital files for scanned manuscripts and other media. Backend services were implemented in C, Java 8, and Scala. Structured catalogue records were maintained in relational databases, while large digital collection files were stored in AWS S3 object storage.

Project roles
Software Engineer

Period
01.2015 – 12.2017

Responsibilities

- Developed and maintained backend modules in C for library catalogue management and digital collection workflows, focusing on data processing, validation, and core business logic;
- Designed and optimized socket-based communication components in C and Java to support concurrent client sessions and data exchange between distributed backend services;
- Extended service APIs and Scala endpoints to support evolving catalogue workflows and integration with existing application modules;
- Integrated AWS S3 object storage through the S3 API to persist digital collection assets, including scanned manuscripts and other large media files, while maintaining bibliographic metadata and object references in relational databases;
- Implemented streaming upload and byte-range download mechanisms for large digital objects, reducing memory consumption and avoiding unnecessary loading of complete files into application memory;
- Updated MySQL and PostgreSQL schemas and optimized database queries to improve data retrieval performance when handling large library catalogues;
- Improved backend data-processing workflows and maintained compatibility with existing legacy application components;
- Participated in release preparation, deployment workflow improvements, and code reviews to maintain application stability and code quality.

Environment
C, Java, Scala, Linux, GDB, MySQL, PostgreSQL, S3 API, Git, Bitbucket, AWS (S3).

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CORPORATE SYSTEM FOR THE OIL AND GAS INDUSTRY

An internal corporate information system for the oil and gas industry, used across departments and operational centers. The backend was developed in C++11, with a web interface built using JavaScript, HTML, and jQuery. Microsoft SQL Server stored structured business records, while LevelDB provided embedded, disk-backed key-value storage for selected high-frequency telemetry and event data. The application ran on Linux servers, with Nginx supporting web access.

Project roles
Software Engineer

Period
08.2012 – 12.2014

Responsibilities

- Developed and maintained backend components in C++11 for internal corporate workflows and operational data processing in the oil and gas domain;
- Integrated LevelDB as an embedded, disk-backed key-value storage engine into C++ backend services to persist and index high-frequency telemetry and sensor events locally on disk, complementing relational storage in Microsoft SQL Server;
- Optimized LevelDB storage performance by tuning block cache configuration and compaction settings to improve write throughput and maintain efficient data access under sustained ingestion workloads;
- Developed and maintained database access layers, supporting interaction between embedded key-value storage and relational database systems;
- Maintained MySQL persistence components for auxiliary operational data and application metadata, while Microsoft SQL Server remained the primary relational database for corporate business records;
- Supported integration between backend services and the web interface implemented using JavaScript, HTML, and jQuery;
- Refactored existing C++ components, resolved defects, and removed outdated code to improve application reliability and maintainability;
- Participated in technical task estimation, code reviews, and development planning to support predictable delivery of project functionality.

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Environment

C++11, JavaScript, HTML, Linux, Visual Assist, CMake, Visual Studio, LevelDB, jQuery, MySQL, Microsoft SQL Server, Nginx, Git, Bitbucket.

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Professional skills

SKILLS		EXPERIENCE IN YEARS	LAST USED
Programming Languages	C++	11	2026
	C	3	2017
	C#	3	2026
	Python	2	2024
	Java	3	2017
	JavaScript	5	2026
	Ruby	2	2024
	MATLAB	4	2022
	Scala	3	2017
	HTML	2	2014
Operating Systems	Linux	5	2017
	Embedded Linux	3	2026
Tools	Kubernetes	2	2024
	Yocto Project	3	2026
	Grafana	2	2024
	Prometheus	2	2024
	Ceph RADOS	2	2024
	Ceph CSI	2	2024
	Qt Designer	3	2026
	Qt Designer .ui forms	3	2026
	RBD	2	2024
	System Tool Kit	2	2020
	systemd	3	2026
	Rook Operator	2	2024
	Visual Assist	2	2014
	CMake	7	2026

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SKILLS		EXPERIENCE IN YEARS	LAST USED
	qmake	3	2026
	gcc	3	2026
	GDB	6	2026
	Valgrind	3	2026
	Qt Test	3	2026
	Pytest	2	2024
	Visual Studio	4	2020
	Qt Creator	3	2026
Frameworks & Libraries	GStreamer	3	2026
	Qt	3	2026
	Qt Widgets	3	2026
	QSS	3	2026
	Ruby on Rails	2	2024
	.NET	3	2026
	Node.js	3	2026
	WinForms	3	2026
	LevelDB	2	2014
	V4L2	3	2026
	jQuery	2	2014
	Qt DBus	3	2026
	Qt WebSockets	3	2026
Multimedia	WebRTC	3	2026
	Qt Multimedia	3	2026
	Linphone-SDK	3	2026
	liblinphone	3	2026
	mediasoup	3	2026
	libmediasoupclient	3	2026

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SKILLS		EXPERIENCE IN YEARS	LAST USED
Databases	MySQL	5	2017
	PostgreSQL	9	2024
	SQLite	3	2026
	LevelDB	2	2014
	Microsoft SQL Server	2	2014
Network	SDP	3	2026
	Nginx	2	2014
	SIP	3	2026
	RTP	3	2026
	SNMP	3	2026
	DHCP	3	2026
	TFTP	3	2026
Cloud	AWS	3	2017
Embedded	BitBake	3	2026
	NXP i.MX	3	2026
Development Processes	Jira	2	2020
	Asana	3	2026
Source control systems	Git	14	2026
	GitLab	3	2026
	Bitbucket	7	2022
Architecture	Qt Model/View	3	2026

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